

11. Gaspereaux-Panmure Island

Shoreline length: 28 km



Shore Types

This SMU includes Gaspereaux shores, Panmure Island and its causeway. Main shore types are dunes, low plains, wetlands, and cliffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
39	5	4	13	19	0	20	0



Causeway and Provincial Park to the right



Armouring along cliff updrift of Panmure Island Provincial Park.



Bluff at Panmure Island Lighthouse



Eroding bluff feeding a beach on north shore of Panmure Island

Governance, Infrastructure, and Economy

Gaspereaux-Panmure Causeway is entirely within the Rural Municipality of Three Rivers. The causeway was built on a barrier dune and provides the sole access route to Panmure Island and is managed by PEI DTI who have conducted local coastal engineering maintenance work since 2005. Infrastructure includes Reilly's Road (Gaspereaux), Panmure Island Road (Route 347, Gaspereaux), the Panmure Head Lighthouse, and Route 347 (Chemin Panmure Island). Local and legacy land use constraints include several small lots and several unpaved roads, including the causeway itself; the single access road is a flood hazard risk, a special consideration, since it is the only route on or off Panmure Island.

Coastal Flood and Erosion Risk

The causeway is at risk of flooding throughout, and under erosion pressure along its north half. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Natural Assets and Heritage

The area includes the Panmure Island Lighthouse, and Panmure Island Cultural Grounds, significant as the site of the annual Abegweit Pow Wow. Panmure Island Provincial Park beach is located at the south end of the causeway. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Planning and Policy Framework

Three Rivers has municipal planning authority. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sediment transport

Net sediment transport is high along the outer coast, where it converges toward the Provincial Park beach at the south end of the causeway. It is moderate around Panmure Islands, where it feeds beaches at Billhook Point and Marsh Point.

Estimated number of buildings and length of road at risk [per km shoreline]

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	11 [<1/km]	15 [1/km]	22 [1/km]	31 [1/km]	4 [<1/km]	20 [1/km]	39 [1/km]
Roads	2608 m [93 m/km]	3371 m [120 m/km]	3969 m [141 m/km]	5039 m [180 m/km]	1349 m [48 m/km]	2238 m [80 m/km]	2694 m [96 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
11.00 Gaspereaux-Panmure Island	NAI	MR	MR
Rationale	It is essential that sediment supply is maintained to the causeway beach, both to preserve the recreational value of the Provincial Park beach, as well as the storm protection value for the causeway link itself. This would be achieved by a strategy of No Active Intervention updrift of the causeway, followed by Managed Realignment where assets are at risk including but not limited to the Lighthouse at Panmure Head.		
11.01 Panmure Island causeway -PU	R	R	R
Rationale	The causeway is a critical link to the Panmure Island community. Resist strategy must incorporate sand nourishment to sustain Provincial Beach Park.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.